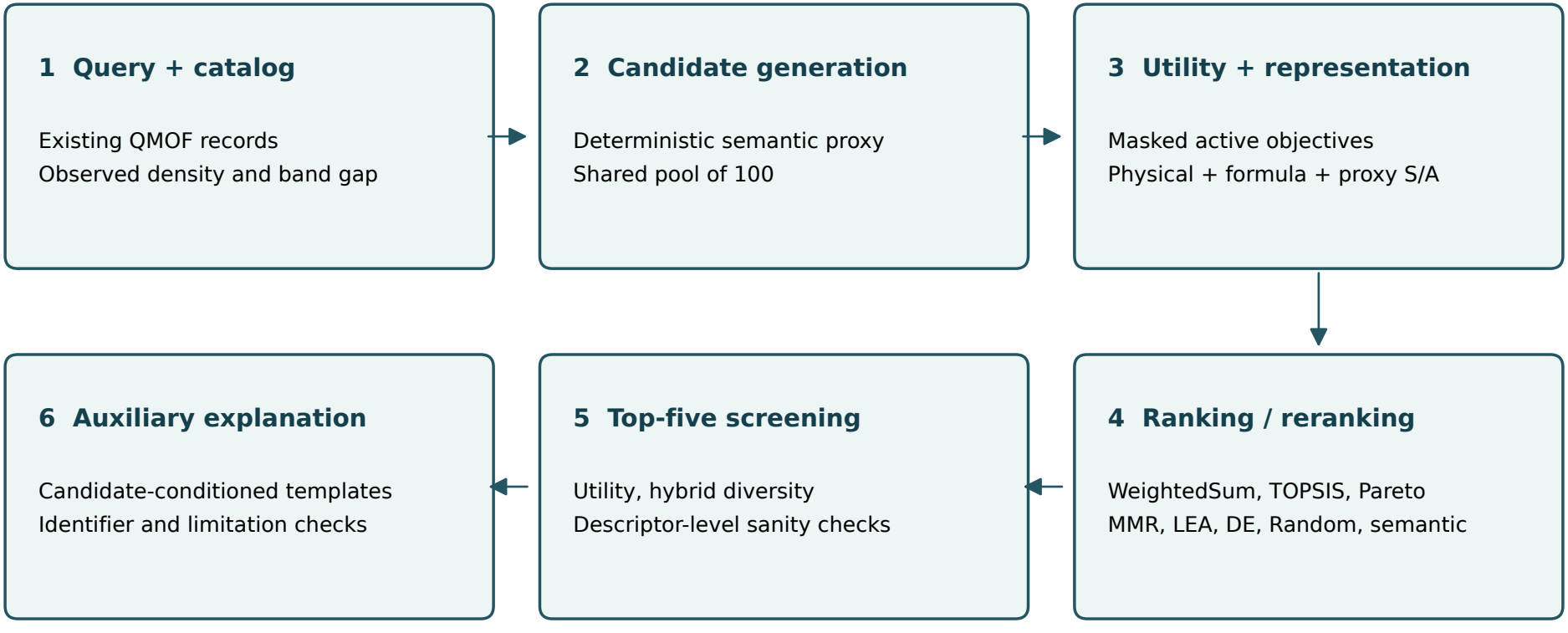
EXECUTED OFFLINE BENCHMARK



Evidence boundaries

Void fraction excluded; physical gap is masked; proxy A retains its zero fallback.

No trained GNN, independent LLM-quality benchmark, or application-performance validation.

PROSPECTIVE: learned GraphSAGE/GAT or CIF-derived representations; independent expert/physical validation.